SMART INDIA HACKATHON 2026

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- **Problem Statement ID –SIH26133**
- **Problem Statement Title- VitalSense**
- **Theme- MedTech/BioTech/ HealthTech**
- **PS Category- Software**
- **Team ID- 142128**
- **Team Name- AarogyaX**



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VitalSense

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The Problem

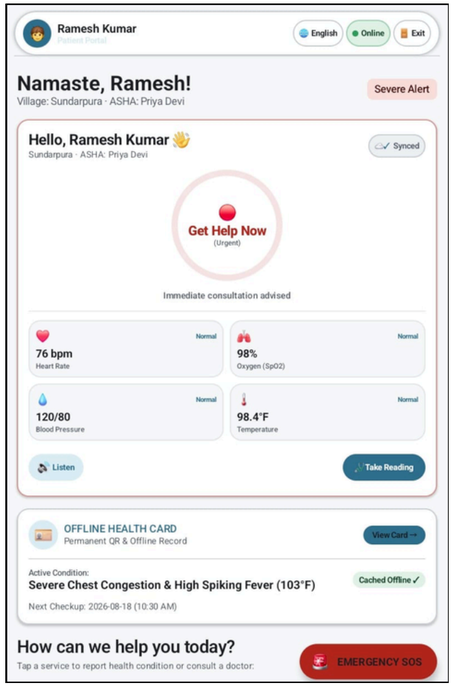
Rural patients often struggle to get timely healthcare because of long travel distances, limited specialist access, poor internet connectivity, and fragmented health records. Manual appointments and weak referral tracking can lead to long waiting times and delayed treatment. At the same time, health officials may lack centralized village-level data to properly monitor and manage community healthcare needs.

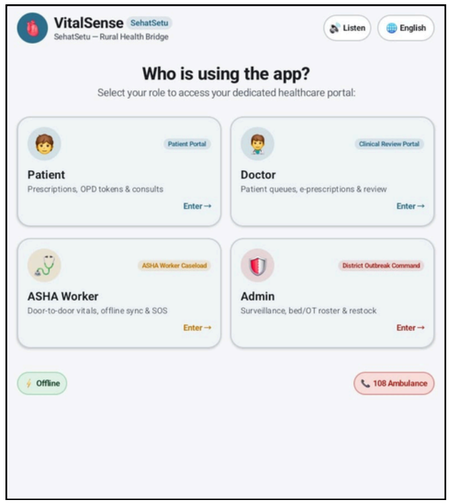
Our Solution

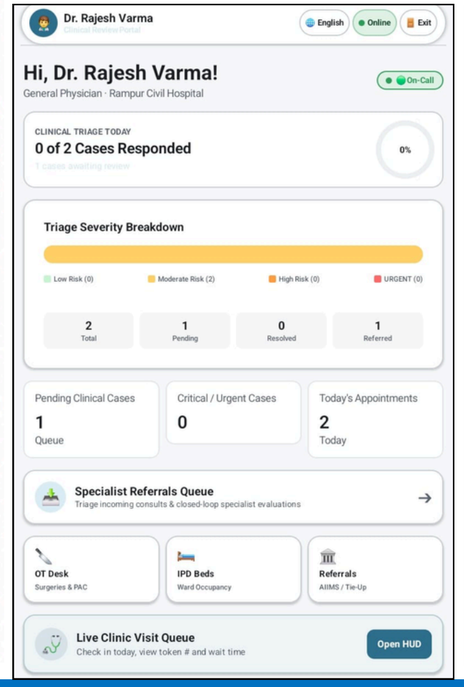
- **One platform for 4 roles** – Connects Patients, ASHA Workers, Doctors & District Admins.
- **Offline-first healthcare** – Patient registration, surveys, vitals and records work without internet and sync automatically when connectivity returns.
- **Digital triage** – Uses vitals and symptoms to identify high-risk patients and enable faster escalation.
- **Teleconsultation** – Enables remote doctor consultations, with vitals visible during the consultation.
- **Closed-loop referrals** – Doctors can refer patients to specialists and track referral status until completion.
- **Digital health records & OCR** – Converts paper prescriptions into structured digital records using on-device OCR.
- **Emergency SOS** – Provides 108 ambulance calling + ASHA SMS/GPS fallback when internet is unavailable.
- **Disease surveillance** – District admins can monitor village-level disease trends and outbreak clusters.
- **Simple & accessible** – Supports multiple languages, proxy care through ASHA workers, and field-friendly interfaces.

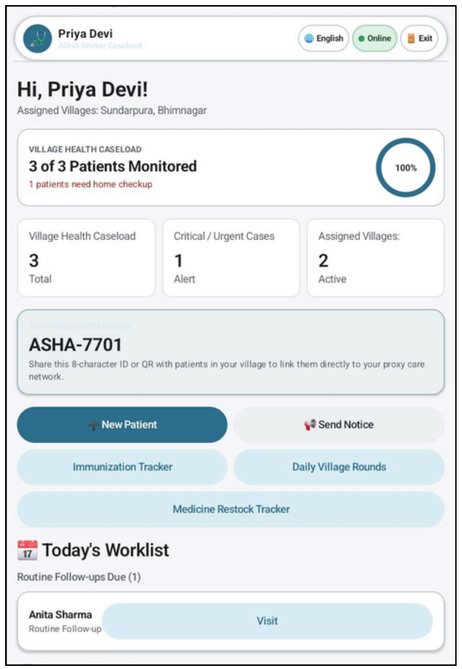
Uniqueness

FEATURES	OTHER APPS / SOLUTIONS	OUR APP (UNIQUE)
Teleconsultation	✗	✓
Appointments & Queue	✗	✓
Digital Triage	✗	✓
Health Records	✗	✓
Referral Tracking	✗	✓
Diagnostics Coordination	✗	✓
Medicine Availability	✗	✓
High-Risk Follow-up	✗	✓
Dashboards	−	✓
Offline-First	−	✓
Secure & Interoperable	−	✓
Emergency Escalation (SOS)	✗	✓
CHW / ASHA Proxy Care	✗	✓







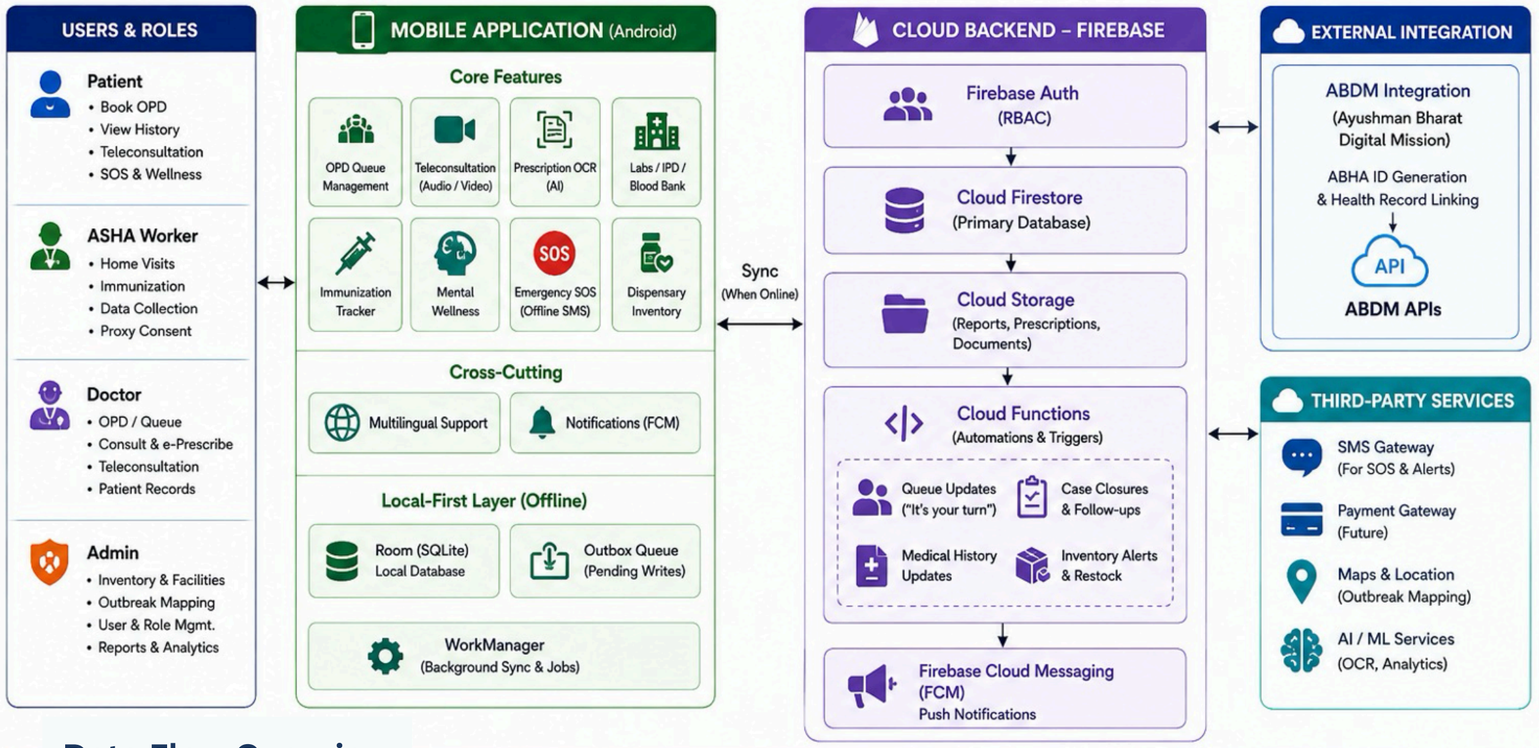


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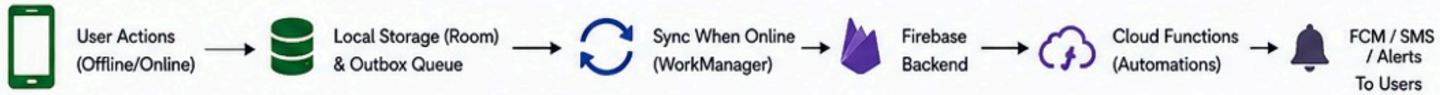
TECHNICAL APPROACH

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ARCHITECTURE



Data Flow Overview



Live Working Prototype & Verification

- *Scan the QR code to access the active demonstration build and compiled Android application package (APK).*
- *Hardware validated: Tested successfully on Android 8.0+ devices with 2GB RAM under simulated zero-connectivity conditions.*

METHODOLOGY

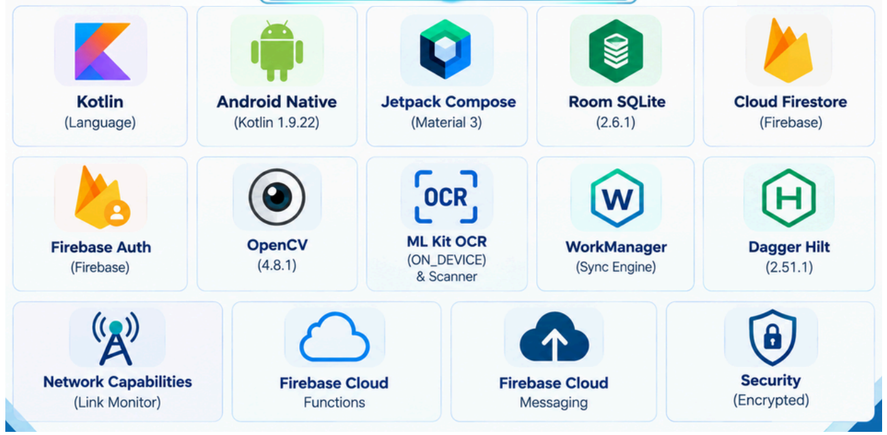
1. Client Ingestion & Local Store: Frontline interactions (OPD triage, vitals, prescriptions) commit locally to Room SQLite within an atomic transaction; offline outbox record created.

2. Adaptive Synchronization: WorkManager evaluates network connectivity; triggers idempotent HTTPS batch synchronization to Cloud Firestore upon signal acquisition.

3. Serverless Event Automations: Cloud Functions execute authoritative business logic: M/M/1 OPD token assignment, inventory decrement triggers, and case escalation listeners.

4. Multi-Channel Alert Dispatch: FCM dispatches real-time push notifications to online clients; cellular SMS Gateway fallback triggers automatically for critical emergency alerts.

TECH STACK



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FEASIBILITY AND VIABILITY

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Comprehensive Feasibility Assessment (Technical, Operational & Economic):

Technical Feasibility & Hardware Fit:

- **Hardware Compatibility**: Runs on 2GB RAM, Android 8.0+ devices, including ASHA tablets..
- **Zero Cloud Cost**: ML Kit OCR runs on-device, requiring no mobile data or cloud API costs for triage.
- **Cloud Scalability**: Firebase serverless automatically scales for peak OPD loads without dedicated servers.

Operational & Clinical Workflow Fit:

- **NHM Alignment**: Triage, maternal tracking and referrals follow existing Maharashtra health workflows
- **ASHA Integration**: Dual-mode UI supports both self-registration and ASHA-assisted workflows.
- **Interoperability**: Supports ABDM standards for integration with national health registries.

Economic & Regulatory Viability:

- **Zero Licensing Cost**: Uses open-source technologies like Kotlin, SQLite, WebRTC and Jetpack.
- **Data Privacy**: Uses AES-256 local encryption and follows applicable data-protection principles..
- **Low Operating Cost**: Serverless, pay-per-use infrastructure minimises idle cloud costs.

Identified Risk & Challenge	Severity & Impact	Root Failure Cause	Engineered Architectural Mitigation in VitalSense
1. OCR Prescription Errors	High (Clinical Error)	Poor camera focus, wrinkled slips, and illegible doctor handwriting.	On-device confidence score thresholding (<85% flagged for manual confirmation); mandatory doctor/pharmacist digital sign-off; original prescription scan permanently attached to patient record.
2. Poor Rural Connectivity	High (Service Outage)	Unstable 2G/3G network towers and remote rural dead zones.	Local-first Room SQLite persistent database with Transactional Outbox pattern; mutations committed locally and synced via WorkManager background retry queues with exponential backoff.
3. Emergency in Zero Network	Critical (Life Safety)	Complete lack of mobile internet data during acute trauma/obstetric events.	Dual-channel emergency fallback: if IP transmission fails within 5 seconds, the application automatically constructs and dispatches an encrypted cellular SMS containing GPS coordinates to 108 and the PHC.
4. Data Privacy & Intermediary Misuse	High (Data Breach)	Shared ASHA tablet devices and unauthorized access to medical records.	Local SQLite database encryption using SQLCipher (AES-256); Firebase granular Role-Based Access Control (RBAC); ABDM-compliant electronic consent flow before sharing longitudinal medical records.
5. Low Digital Literacy	Medium (Low Adoption)	Complex multi-tier UI layouts and English technical medical terminology.	Multilingual vernacular localisation (Marathi, Hindi, English); high-contrast icon-driven workflows; audio-guided prompts; dedicated ASHA-assisted proxy workflow for illiterate citizens.
6. Limited Outbreak Data	Medium (Delayed Action)	Batch-delayed or incomplete manual paperwork masking early epidemic clusters.	Automated syndromic case tallying at point of care; automated geospatial spatial-temporal clustering (DBSCAN); push notifications sent to Taluka Health Officers upon exceeding statistical thresholds.

Scalability, Public Health Deployment & Government Sustainability Road map:

Administrative Tier Alignment:

Fully compatible with the 4-tier public health hierarchy of Maharashtra: Sub-Centre (ASHA/ANM) -> Primary Health Centre (Medical Officer) -> Community Health Centre (Specialists) -> District Hospital (Tertiary Referral).

Infrastructure Cost Efficiency:

Pay-as-you-go serverless backend incurs zero idle compute costs; local-first processing reduces mobile internet bandwidth demand by >75% compared to conventional web dashboards.

Deployment Road map:

Phase 1: Pilot validation across 10 rural PHCs and 50 Sub-centres; Phase 2: Taluka-level integration with district hospital referral registries; Phase 3: State-wide expansion under Maharashtra DHS.

Frontline Capacity Building:

Intuitive icon-driven vernacular interfaces (Marathi/Hindi) require under 2 hours of training for ASHA cadres, ensuring rapid field adoption with zero disruption to daily healthcare delivery.

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IMPACT AND BENEFITS

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Potential Impact on Target Audience

OPD Registration & Patient Waiting

Target: ~75 min → <5 min

Digital registration and token management can reduce queues at PHCs, helping patients spend less time waiting and allowing staff to handle registrations more efficiently.

Patient Travel & Out-of-Pocket Costs

Target: 30–70 km and ₹250–₹500 saved/visit

Teleconsultation and digital prescription/follow-up support can reduce avoidable trips to distant hospitals, particularly for routine consultations and follow-ups.

Referral & Specialist Access

Target: 7–14 days → <24 hrs

Digital referrals allow PHC staff to share patient information with higher centres and track appointment confirmation, reducing delays in specialist care.

Early Community Health Alerts

Target: <24 hrs instead of days/weeks

Aggregated symptom and case reporting can help health workers identify unusual fever, diarrhea or other disease clusters earlier and alert the relevant health authorities.

ASHA Worker Efficiency

ASHA workers can use the system to register patients, update records, assist with referrals and monitor follow-ups from a single workflow, reducing dependence on scattered paper-based processes

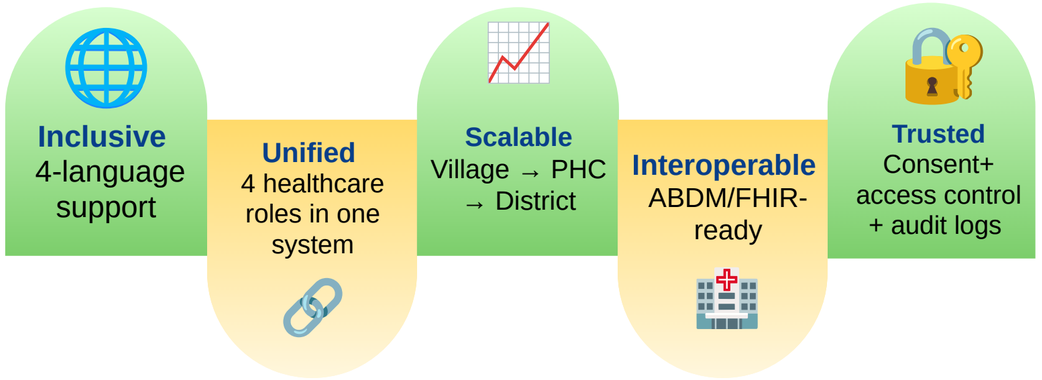
Continuity of Patient Care

A digital patient record gives authorized healthcare workers access to previous visits, referrals, prescriptions and follow-up information, reducing loss of information when a patient moves between healthcare facilities.

Rural Patient Accessibility

The system reduces dependence on distance, specialist availability and repeated physical visits, giving underserved patients a more connected path from PHC → specialist → follow-up.

Long Term Value



BENEFITS

SOCIAL

- **Improved Rural Healthcare Access** - Connects remote villages to doctors and health services.
- **Empowers ASHA Workers** - Lets health workers manage care for patients without smartphones.
- **Faster Emergency Response** - Auto-escalates calls, falls back to SMS+GPS if unanswered.

ECONOMIC

- **Reduced Waiting & Travel Time** - Real-time queue tracking cuts unnecessary clinic visits.
- **Medicine Availability** - Shows free/low-cost pharmacy options like Jan Aushadhi stores.
- **Cost Savings -** Early detection and teleconsultation reduce treatment delays and costs.

OPERATIONAL

- **Offline-First Healthcare** - Works without internet, syncs automatically once connected.
- **Digital Health Records** - OCR-scanned prescriptions replace easily-lost paper records.
- **Smart Queue Management** - Real-time ETAs and prevents doctor double-booking.

ENVIRONMENTAL

- **Less Paper Usage** - Digital prescriptions and records reduce paper dependency
- **Reduced Travel & Emissions** - Teleconsultation cuts unnecessary hospital trips

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RESEARCH AND REFERENCES

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Reseach paper	Methodology	Reseach Gap	Our proposed work
ABDM Architecture Assessment Neogi, S. B., Pagaria, V., Dutta, S., Bele, N., Yadav, P., & Samtani, R. (2024). Scan and share to register outpatients, India. Bulletin of the World Health Organization, 103(2), 164.	Policy & technical analysis of NHA data & M1–M3 adoption across 35 states/UTs	Cloud-dependent APIs fail during rural blackouts; exchanges static PDFs without computable triage data.	Offline-First Room SQLite Gateway: Local caching, discrete FHIR data schema, and DPDP-compliant proxy consent.
ASHA mHealth cRCT Nagraj, S., Kennedy, S., Jha, V., Norton, R., Hinton, L., Billot, L., ... & Hirst, J. (2023). A mobile clinical decision support system for high-risk pregnant women in rural India (SMARThealth pregnancy): pilot cluster randomized controlled trial. JMIR Formative Research, 7, e44362.	Cluster RCT across 4 rural PHCs evaluating an Android tablet CDSS for antenatal screening.	No closed-loop confirmation when patients reach hospital; siloed to maternal care, ignoring child/acute triage.	Closed-Loop Referral Tracking: Bi-directional doctor confirmation loop + unified multi-condition triage engine.
ABDM OPD Tokenisation Neogi, S. B., Pagaria, V., Dutta, S., Bele, N., Yadav, P., & Samtani, R. (2024). Scan and share to register outpatients, India. Bulletin of the World Health Organization, 103(2), 164.	Cross-sectional study of 16,939 facilities evaluating manual vs. QR Scan & Share tokenisation	Requires patient smartphones and digital literacy; blind sequential tokens still cause waiting room crowding.	ASHA Proxy QR Tokens: Assisted registration + dynamic M/M/1M/M/1 queue estimation calculating exact arrival times
CHW Task-Shifting mHealth *Bassi, A., Arfin, S., John, O., Praveen, D., Arora, V., Kalra, O. P., ... & Jha, V. (2023). Innovative mobile-health led participatory approach to comprehensive screening and treatment of diabetes (IMPACT diabetes): rationale, design, and baseline characteristics. International Journal of Diabetes in Developing Countries, 43(3), 353-362.*	cRCT across 16 rural clusters evaluating ASHA-led CBAC screening and point-of-care testing	Doctor cognitive overload from raw unprioritized alerts; unlinked to PHC drug stock; complex text forms.	3-Tier Pre-Triage Filtering: Automatic Green/Amber/Red tagging + real-time dispensary inventory linking + icon UI.
Local-First Tree Sync Kleppmann, M., Mulligan, D. P., Gomes, V. B., & Beresford, A. R. (2021). A highly-available move operation for replicated trees. IEEE Transactions on Parallel and Distributed Systems, 33(7), 1711-1724.	Formal verification of CRDT tree algorithm using Isabelle/HOL for coordination-free synchronization.	Theoretical model unoptimized for low-cost Android RAM/battery; ignores medical legal immutability rules	Room SQLite Outbox Pattern: Production mobile sync with strict append-only clinical records & WorkManager scheduling.
Prescription ML OCR Hassan, E., Tarek, H., Hazem, M., Bahnacy, S., Shaheen, L., & Elashmwai, W. H. (2021, January). Medical prescription recognition using machine learning. In 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC) (pp. 0973-0979). IEEE.	Computer vision preprocessing + CNN-LSTM neural network for handwritten prescription recognition.	High clinical liability from unverified misreads; heavy models require cloud GPUs, failing offline.	On-Device Google ML Kit OCR: Zero cloud cost/data + mandatory <85% <85% confidence doctor verification and scan archiving.
Rural Telemedicine Dastidar, B. G., Jani, A. R., Suri, S., & Nagaraja, V. H. (2024). Reimagining India's National Telemedicine Service to improve access to care. The Lancet Regional Health-Southeast Asia, 30.	National evaluation of eSanjeevani (>200M calls) comparing provider hub-and-spoke vs. direct OPD.	Video calls collapse on rural 2G/3G; zero pre-consultation vitals; prescriptions disconnected from local stock.	Adaptive WebRTC Fallback: Seamless downgrade to 32 kbps audio on weak signals + pre-loaded vitals + Cellular SMS SOS.